STAR RX72N - blinky_rtos Bench Test

1.0.0

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Chapter 1

Directory Hierarchy

1.1 Directories

Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

Chapter 3

Directory Documentation

3.1 blinky_rtos Directory Reference

Files

- file linker.ld
- file main.c

  RX72N multi-LED breathing cycle using ThreadX RTOS.

Chapter 4

File Documentation

4.1 blinky_rtos/linker.ld File Reference

4.2 linker.ld

Go to the documentation of this file.

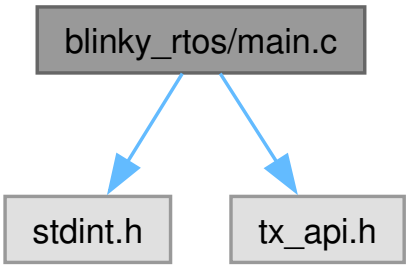
```
00001 /*
00002  * blinky_rtos/linker.ld -- Linker script for ThreadX-based multi-LED breathe.
00003  *
00004  * RX72N memory map (R5F572NNHxFB):
00005  *   Internal RAM  : 0x00000004 - 0x0007FFFF  (512 KB, first 4 bytes reserved)
00006  *   Internal ROM  : 0xFFE00000 - 0xFFFFFFFF  (2 MB)
00007  *
00008  * Fixed hardware vector locations:
00009  *   0xFFFFFF80 - 0xFFFFFFFC  Exception vector table (EXTB)
00010  *   0xFFFFFFFC - 0xFFFFFFFF  Reset vector (points to startup entry)
00011  *
00012  * INTB is loaded by startup.S to point at _rvectors_start (in ROM).
00013  *
00014  * Renesas-syntax sections from ThreadX RXv3 port (tx_timer_interrupt.S etc.):
00015  *   P, P_1  -- code sections (.SECTION P,CODE)
00016  *   B, B_1  -- BSS sections  (.SECTION B,DATA)
00017  *   D, D_1  -- initialised data sections
00018  */
00019
00020 MEMORY
00021 {
00022     RAM : ORIGIN = 0x00000004, LENGTH = 0x7FFFC   /* 512 KB - 4 bytes */
00023     ROM : ORIGIN = 0xFFE00000, LENGTH = 0x200000  /* 2 MB             */
00024 }
00025
00026 SECTIONS
00027 {
00028     /* Code -- explicit ROM base so section VMA is anchored correctly.
00029      * *(P) captures Renesas-assembler code sections from ThreadX port files. */
00030     .text 0xFFE00000 : AT(0xFFE00000)
00031     {
00032         *(.text*)
00033         *(P)
00034         *(P_1)
00035         *(.rodata*)
00036         *(C)
00037         *(C_1)
00038         *(C_2)
00039         etext = .;
00040     } > ROM
00041
00042     /*
00043      * Relocatable vector table -- startup.S loads INTB with this address.
00044      * Must be 4-byte aligned; 32 entries = 128 bytes.
00045      * No AT() needed: follows .text in ROM, VMA and LMA are the same.
00046      */
00047     .rvectors ALIGN(4) :
00048     {
00049         _rvectors_start = .;
```

```
00050         KEEP(*(.rvectors))
00051         _rvectors_end = .;
00052     } > ROM
00053
00054     /*
00055      * Initialised data -- LMA stays in ROM (image stored here), VMA in RAM.
00056      * startup.S copies from _mdata (ROM) to _data..._edata (RAM) on boot.
00057      * Required for ThreadX: ___tx_thread_system_state starts as 0xF0F0F0F0.
00058      */
00059     _mdata = .;
00060     .data : AT(_mdata)
00061     {
00062         _data = .;
00063         *(.data*)
00064         *(D)
00065         *(D_1)
00066         *(D_2)
00067         _edata = .;
00068     } > RAM
00069
00070     /* BSS -- ThreadX needs this zeroed; startup.S handles it.
00071      * B/B_1/B_2 are Renesas-assembler BSS sections from ThreadX port files. */
00072     .bss :
00073     {
00074         _bss = .;
00075         *(.bss*)
00076         *(COMMON)
00077         *(B)
00078         *(B_1)
00079         *(B_2)
00080         _ebss = .;
00081         . = ALIGN(4);
00082         _end = .;           /* ThreadX tx_initialize_low_level reads _end */
00083     } > RAM AT>RAM
00084
00085     /*
00086      * Stacks -- USP grows down from top of RAM, ISP 2 KB below that.
00087      * ThreadX uses the interrupt stack pointer (ISP) for ISR execution.
00088      */
00089     _ustack = ORIGIN(RAM) + LENGTH(RAM);
00090     _istack = _ustack - 0x800;
00091
00092     /*
00093      * Exception vector table (EXTB) at fixed hardware address 0xFFFFFF80.
00094      */
00095     .exvectors 0xFFFFFF80 : AT(0xFFFFFF80)
00096     {
00097         LONG(0xFFFFFFFF);   /* BUSERR */
00098         LONG(0xFFFFFFFF);   /* reserved */
00099         LONG(0xFFFFFFFF);   /* FIFERR */
00100         LONG(0xFFFFFFFF);   /* FRDYI */
00101         LONG(0xFFFFFFFF);   /* reserved */
00102         LONG(0xFFFFFFFF);
00103         LONG(0xFFFFFFFF);
00104         LONG(0xFFFFFFFF);
00105         LONG(0xFFFFFFFF);
00106         LONG(0xFFFFFFFF);
00107         LONG(0xFFFFFFFF);
00108         LONG(0xFFFFFFFF);
00109         LONG(0xFFFFFFFF);
00110         LONG(0xFFFFFFFF);
00111         LONG(0xFFFFFFFF);
00112         LONG(0xFFFFFFFF);
00113         LONG(0xFFFFFFFF);
00114         LONG(0xFFFFFFFF);
00115         LONG(0xFFFFFFFF);
00116         LONG(0xFFFFFFFF);
00117         LONG(0xFFFFFFFF);
00118         LONG(0xFFFFFFFF);
00119         LONG(0xFFFFFFFF);
00120         LONG(0xFFFFFFFF);
00121         LONG(0xFFFFFFFF);
00122         LONG(0xFFFFFFFF);
00123         LONG(0xFFFFFFFF);
00124         LONG(0xFFFFFFFF);
00125         LONG(0xFFFFFFFF);
00126         LONG(0xFFFFFFFF);
00127         LONG(0xFFFFFFFF);
00128     } > ROM
00129
00130     /* Fixed reset vector at 0xFFFFFFFC */
00131     .fvectors 0xFFFFFFFC : AT(0xFFFFFFFC)
00132     {
00133         KEEP(*(.fvectors))
00134     } > ROM
00135 }
```

4.3 blinky_rtos/main.c File Reference

RX72N multi-LED breathing cycle using ThreadX RTOS.

#include <stdint.h>
#include "tx_api.h"
Include dependency graph for main.c:



Data Structures

- struct led_t

Enumerations

- enum hw_addr_t : uintptr_t { k_port7_pdr_addr = 0x0008C007U , k_port7_podr_addr = 0x0008C027U , k_porta_pdr_addr = 0x0008C00AU , k_porta_podr_addr = 0x0008C02AU , k_portb_pdr_addr = 0x0008C00BU , k_portb_podr_addr = 0x0008C02BU , k_cmstr0_addr = 0x00088000U , k_cmt0_cmcr = 0x00088002U , k_cmt0_cmcnt = 0x00088004U , k_cmt0_cmcor = 0x00088006U , k_icu_ir_base = 0x00087000U , k_icu_ier_base = 0x00087200U , k_icu_ipr_base = 0x00087300U }
- enum porta_pins_t : uint8_t { k_pin_pa7 = (uint8_t)(1U << 7U) }
- enum port7_pins_t : uint8_t { k_pin_p71 = (uint8_t)(1U << 1U) , k_pin_p72 = (uint8_t)(1U << 2U) }
- enum portb_pins_t : uint8_t { k_pin_pb0 = (uint8_t)(1U << 0U) , k_pin_pb1 = (uint8_t)(1U << 1U) , k_pin_pb2 = (uint8_t)(1U << 2U) }
- enum cmt0_cfg_t : uint16_t { k_cmt0_cmcor_val = 299U , k_cmt0_cmcr_val = 0x40U , k_cmstr0_ch0_bit = 0x01U }
- enum icu_cmt0_cfg_t : uint8_t { k_vect_cmt0 = 28U , k_cmt0_priority = 3U , k_icu_ier_cmt0 = 3U , k_icu_ier_cmt0_bit = 4U }
- enum breathe_params_t : uint32_t { k_max_bright = 50U , k_reps_per_step = 5U , k_flash_count = 3U , k_flash_on_iters = 4000U , k_flash_off_iters = 2000U }
- enum rtos_prio_t : uint8_t { k_num_leds = 6U , k_breathe_priority = 10U , k_flash_priority = 9U }
- enum rtos_stack_t : uint16_t { k_breathe_stack_sz = 512U , k_flash_stack_sz = 256U }

Functions

- static volatile uint8_t * port7_pdr (void)
- static volatile uint8_t * port7_podr (void)
- static volatile uint8_t * porta_pdr (void)
- static volatile uint8_t * porta_podr (void)
- static volatile uint8_t * portb_pdr (void)
- static volatile uint8_t * portb_podr (void)
- static volatile uint16_t * cmstr0 (void)
- static volatile uint16_t * cmt0_cmcr (void)
- static volatile uint16_t * cmt0_cmcnt (void)
- static volatile uint16_t * cmt0_cmcor (void)
- static volatile uint8_t * icu_ir (uint8_t vect)
- static volatile uint8_t * icu_ier (uint8_t idx)
- static volatile uint8_t * icu_ipr (uint8_t vect)
- void _tx_timer_interrupt (void)
- void cmt0_isr (void)
- static void cmt0_init (void)
- static void led_on (const led_t *led)
- static void led_off (const led_t *led)
- static void pwm_delay (uint32_t count)
- static void breathe_ramp (const led_t *led, uint32_t invert)
- static void breathe_task_entry (ULONG arg)
- static void flash_task_entry (ULONG arg)
- void tx_application_define (void *first_unused_memory)
- int main (void)

Variables

- static TX_THREAD s_breathe_tcb
- static TX_THREAD s_flash_tcb
- static TX_SEMAPHORE s_cycle_done_sem
- static TX_SEMAPHORE s_flash_done_sem
- static uint8_t s_breathe_stack [k_breathe_stack_sz]
- static uint8_t s_flash_stack [k_flash_stack_sz]
- static led_t s_leds [k_num_leds]

4.3.1 Detailed Description

RX72N multi-LED breathing cycle using ThreadX RTOS.

Two ThreadX threads drive the breathing animation:

breathe_task – cycles through all 6 LEDs sequentially, applying a software-PWM breathing effect to each one in turn. After completing a full cycle it posts a semaphore.

flash_task – waits on the semaphore, runs a wave sequence: each LED lights up one at a time in order (k_flash_count passes), then signals breathe_task to start the next cycle.

Threading replaces the single infinite loop of the bare-metal version. Busy-wait delays are used for both PWM and wave timing; the 100 Hz tick is too coarse for smooth dimming and tx_thread_sleep is not used.

Clock: LOCO default at 240 kHz (OFS1 = 0xFFFFFFFF, factory reset state). Tick: CMT0 at PCLKB/8 = 30 kHz, CMCOR = 299 -> 100 Hz (10 ms/tick).

Port register addresses: RX72N Hardware Manual r01uh0824ej0111, Ch 22. PDR base = 0x0008C000 + port_index PODR base = 0x0008C020 + port_index Port indices: 7=0x07, A=0x0A, B=0x0B

ICU and CMT0 addresses: RX72N Hardware Manual, Ch 13/15.

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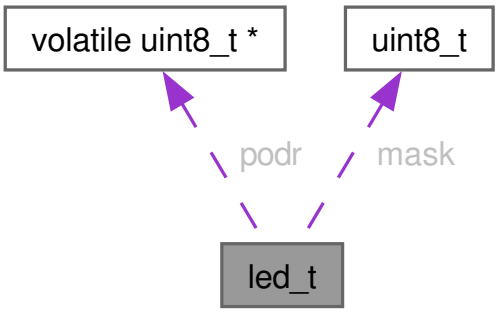
Definition in file main.c.

4.3.2 Data Structure Documentation

4.3.2.1 struct led_t

Definition at line 139 of file main.c.

Collaboration diagram for led_t:



Data Fields

uint8_t	mask	Bit mask for this LED's pin
volatile uint8_t *	podr	Port output data register

4.3.3 Enumeration Type Documentation

4.3.3.1 breathe_params_t

enum breathe_params_t : uint32_t

Enumerator

k_max_bright	PWM duty-cycle upper bound
k_reps_per_step	PWM cycles per brightness step
k_flash_count	Wave passes at cycle end
k_flash_on_iters	Wave LED ON busy-wait iters (~200 ms at LOCO 240 kHz)
k_flash_off_iters	Wave LED OFF gap busy-wait iters (~100 ms at LOCO 240 kHz)

Definition at line 106 of file main.c.

```
00106             : uint32_t {
00107     k_max_bright      = 50U,    /**< PWM duty-cycle upper bound */
00108     k_reps_per_step   = 5U,     /**< PWM cycles per brightness step */
00109     k_flash_count     = 3U,     /**< Wave passes at cycle end */
00110     k_flash_on_iters  = 4000U,  /**< Wave LED ON busy-wait iters (~200 ms at LOCO 240 kHz) */
00111     k_flash_off_iters = 2000U,  /**< Wave LED OFF gap busy-wait iters (~100 ms at LOCO 240 kHz) */
00112 } breathe_params_t;
```

4.3.3.2 cmt0_cfg_t

enum cmt0_cfg_t : uint16_t

Enumerator

k_cmt0_cmcor_val	Compare match for 100 Hz at LOCO/8
k_cmt0_cmcr_val	CMIE=1, CKS=00 (PCLKB/8)
k_cmstr0_ch0_bit	CMSTR0 bit 0 = CMT0 start/stop

Definition at line 87 of file main.c.

```
00087             : uint16_t {
00088     k_cmt0_cmcor_val  = 299U,  /**< Compare match for 100 Hz at LOCO/8 */
00089     k_cmt0_cmcr_val   = 0x40U, /**< CMIE=1, CKS=00 (PCLKB/8) */
00090     k_cmstr0_ch0_bit  = 0x01U, /**< CMSTR0 bit 0 = CMT0 start/stop */
00091 } cmt0_cfg_t;
```

4.3.3.3 hw_addr_t

enum hw_addr_t : uintptr_t

Enumerator

k_port7_pdr_addr	PORT7 direction register
k_port7_podr_addr	PORT7 output data register
k_porta_pdr_addr	PORTA direction register
k_porta_podr_addr	PORTA output data register
k_portb_pdr_addr	PORTB direction register
k_portb_podr_addr	PORTB output data register
k_cmstr0_addr	CMT start register (shared CH0/CH1)
k_cmt0_cmcr	CMT0 control register
k_cmt0_cmcnt	CMT0 counter
k_cmt0_cmcor	CMT0 compare match register
k_icu_ir_base	IR[] base (IR[n] = base + n)
k_icu_ier_base	IER[] base (IER[n/8] bit n%8)
k_icu_ipr_base	IPR[] base (IPR[n] = base + n)

Definition at line 42 of file main.c.

```
00042             : uintptr_t {
00043     /* GPIO */
00044     k_port7_pdr_addr  = 0x0008C007U, /**< PORT7 direction register */
00045     k_port7_podr_addr = 0x0008C027U, /**< PORT7 output data register */
00046     k_porta_pdr_addr  = 0x0008C00AU, /**< PORTA direction register */
00047     k_porta_podr_addr = 0x0008C02AU, /**< PORTA output data register */
00048     k_portb_pdr_addr  = 0x0008C00BU, /**< PORTB direction register */
00049     k_portb_podr_addr = 0x0008C02BU, /**< PORTB output data register */
00050
00051     /* CMT0 (compare match timer 0) */
00052     k_cmstr0_addr = 0x00088000U, /**< CMT start register (shared CH0/CH1) */
00053     k_cmt0_cmcr   = 0x00088002U, /**< CMT0 control register */
00054     k_cmt0_cmcnt  = 0x00088004U, /**< CMT0 counter */
00055     k_cmt0_cmcor  = 0x00088006U, /**< CMT0 compare match register */
00056
00057     /* ICU (interrupt controller) */
00058     k_icu_ir_base  = 0x00087000U, /**< IR[] base (IR[n] = base + n) */
00059     k_icu_ier_base = 0x00087200U, /**< IER[] base (IER[n/8] bit n%8) */
00060     k_icu_ipr_base = 0x00087300U, /**< IPR[] base (IPR[n] = base + n) */
00061 } hw_addr_t;
```

4.3.3.4 icu_cmt0_cfg_t

enum icu_cmt0_cfg_t : uint8_t

Enumerator

k_vect_cmt0	CMT0 CMI0 interrupt vector number
k_cmt0_priority	ICU interrupt priority for CMT0
k_icu_ier_cmt0	IER register index for vector 28 (28/8)
k_icu_ier_cmt0_bit	Bit position within IER[3] (28%8)

Definition at line 93 of file main.c.

```
00093             : uint8_t {
00094     k_vect_cmt0       = 28U,  /**< CMT0 CMI0 interrupt vector number */
00095     k_cmt0_priority   = 3U,   /**< ICU interrupt priority for CMT0 */
00096     k_icu_ier_cmt0    = 3U,   /**< IER register index for vector 28 (28/8) */
00097     k_icu_ier_cmt0_bit = 4U,  /**< Bit position within IER[3] (28%8) */
00098 } icu_cmt0_cfg_t;
```

4.3.3.5 port7_pins_t

enum port7_pins_t : uint8_t

Enumerator

k_pin_p71	PORT7 pin 1
k_pin_p72	PORT7 pin 2

Definition at line 70 of file main.c.

```
00070             : uint8_t {
00071     k_pin_p71 = (uint8_t)(1U « 1U), /**< PORT7 pin 1 */
00072     k_pin_p72 = (uint8_t)(1U « 2U), /**< PORT7 pin 2 */
00073 } port7_pins_t;
```

4.3.3.6 porta_pins_t

enum porta_pins_t : uint8_t

Enumerator

k_pin_pa7	PORTA pin 7

Definition at line 66 of file main.c.

```
00066             : uint8_t {
00067     k_pin_pa7 = (uint8_t)(1U « 7U), /**< PORTA pin 7 */
00068 } porta_pins_t;
```

4.3.3.7 portb_pins_t

enum portb_pins_t : uint8_t

Enumerator

k_pin_pb0	PORTB pin 0
k_pin_pb1	PORTB pin 1
k_pin_pb2	PORTB pin 2

Definition at line 75 of file main.c.

```
00075             : uint8_t {
00076     k_pin_pb0 = (uint8_t)(1U « 0U), /**< PORTB pin 0 */
00077     k_pin_pb1 = (uint8_t)(1U « 1U), /**< PORTB pin 1 */
00078     k_pin_pb2 = (uint8_t)(1U « 2U), /**< PORTB pin 2 */
00079 } portb_pins_t;
```

4.3.3.8 rtos_prio_t

enum rtos_prio_t : uint8_t

Enumerator

k_num_leds	LEDs in the breathing sequence
k_breathe_priority	breathe_task ThreadX priority
k_flash_priority	flash_task ThreadX priority (runs first)

Definition at line 117 of file main.c.

```
00117             : uint8_t {
00118     k_num_leds        = 6U,  /**< LEDs in the breathing sequence */
00119     k_breathe_priority = 10U, /**< breathe_task ThreadX priority */
00120     k_flash_priority   = 9U,  /**< flash_task ThreadX priority (runs first) */
00121 } rtos_prio_t;
```

4.3.3.9 rtos_stack_t

enum rtos_stack_t : uint16_t

Enumerator

k_breathe_stack_sz	breathe_task stack in bytes
k_flash_stack_sz	flash_task stack in bytes

Definition at line 123 of file main.c.

```
00123             : uint16_t {
00124     k_breathe_stack_sz = 512U, /**< breathe_task stack in bytes */
00125     k_flash_stack_sz   = 256U, /**< flash_task stack in bytes */
00126 } rtos_stack_t;
```

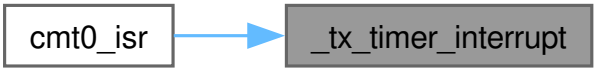
4.3.4 Function Documentation

4.3.4.1 _tx_timer_interrupt()

```
void _tx_timer_interrupt (
                void ) [extern]
```

Referenced by cmt0_isr().

Here is the caller graph for this function:



4.3.4.2 breathe_ramp()

```
void breathe_ramp (
                const led_t * led,
                uint32_t invert) [static]
```

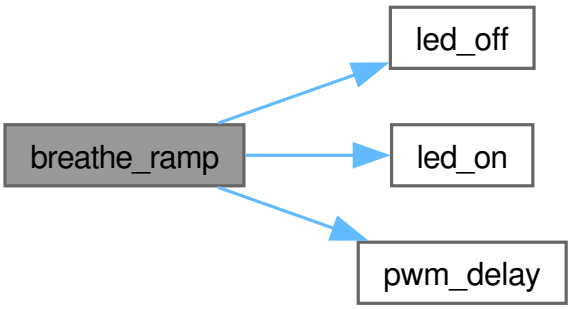
Definition at line 268 of file main.c.

```
00269 {
00270     uint32_t b;
00271     uint32_t rep;
00272     uint32_t on_time;
00273     uint32_t off_time;
00274
00275     for (b = 0U; b <= k_max_bright; b++) {
00276         on_time  = (invert == 0U) ? b : (k_max_bright - b);
00277         off_time = k_max_bright - on_time;
00278         for (rep = 0U; rep < k_reps_per_step; rep++) {
00279             if (on_time > 0U) {
00280                 led_on(led);
00281                 pwm_delay(on_time);
00282             }
00283             led_off(led);
00284             if (off_time > 0U) {
00285                 pwm_delay(off_time);
00286             }
00287         }
00288     }
00289 }
```

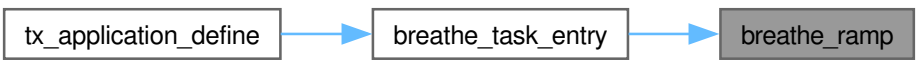
References k_max_bright, k_reps_per_step, led_off(), led_on(), and pwm_delay().

Referenced by breathe_task_entry().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.4.3 breathe_task_entry()

```
void breathe_task_entry (
                ULONG arg)    [static]
```

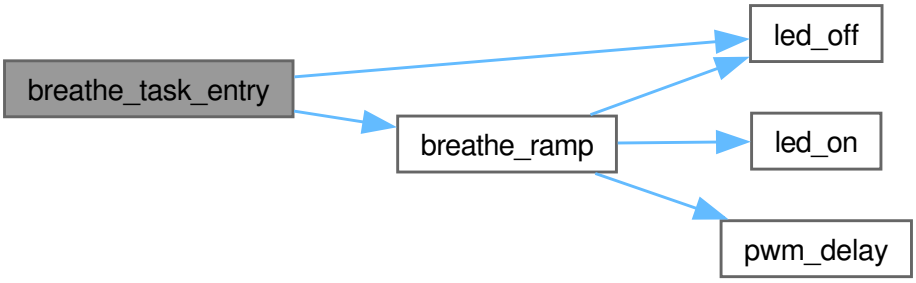
Definition at line 294 of file main.c.

```
00295 {
00296     (void)arg;
00297
00298     for (;;) {
00299         uint8_t i;
00300         for (i = 0U; i < k_num_leds; i++) {
00301             breathe_ramp(&s_leds[i], 0U); /* fade in  */
00302             breathe_ramp(&s_leds[i], 1U); /* fade out */
00303             led_off(&s_leds[i]);
00304         }
00305
00306         /* Signal flash_task that one full cycle is complete */
00307         (void)tx_semaphore_put(&s_cycle_done_sem);
00308
00309         /* Wait for flash_task to finish before starting the next cycle */
00310         (void)tx_semaphore_get(&s_flash_done_sem, TX_WAIT_FOREVER);
00311     }
00312 }
```

References breathe_ramp(), k_num_leds, led_off(), s_cycle_done_sem, s_flash_done_sem, and s_leds.

Referenced by tx_application_define().

Here is the call graph for this function:



Here is the caller graph for this function:

4.3.4.4 cmstr0()

```
volatile uint16_t * cmstr0 (
                void )    [inline], [static]
```

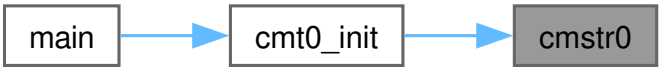
Definition at line 178 of file main.c.

```
00179 {
00180     return (volatile uint16_t *)k_cmstr0_addr;
00181 }
```

References k_cmstr0_addr.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.5 cmt0_cmcnt()

volatile uint16_t * cmt0_cmcnt (
void) [inline], [static]

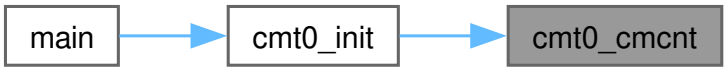
Definition at line 186 of file main.c.

```
00187 {
00188     return (volatile uint16_t *)k_cmt0_cmcnt;
00189 }
```

References k_cmt0_cmcnt.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.6 cmt0_cmcor()

volatile uint16_t * cmt0_cmcor (
void) [inline], [static]

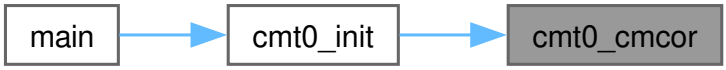
Definition at line 190 of file main.c.

```
00191 {
00192     return (volatile uint16_t *)k_cmt0_cmcor;
00193 }
```

References k_cmt0_cmcor.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.7 cmt0_cmcr()

volatile uint16_t * cmt0_cmcr (
void) [inline], [static]

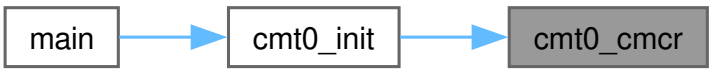
Definition at line 182 of file main.c.

```
00183 {
00184     return (volatile uint16_t *)k_cmt0_cmcr;
00185 }
```

References k_cmt0_cmcr.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.8 cmt0_init()

```
void cmt0_init (
                void ) [static]
```

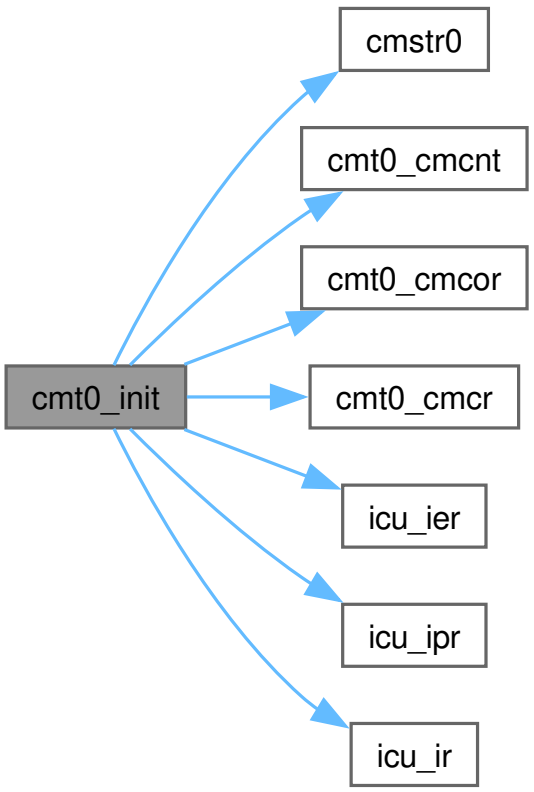
Definition at line 225 of file main.c.

```
00226 {
00227     *cmstr0()    &= (uint16_t)~(uint16_t)k_cmstr0_ch0_bit; /* Stop CMT0 */
00228     *cmt0_cmcr() = k_cmt0_cmcr_val;                        /* PCLKB/8, IRQ enabled */
00229     *cmt0_cmcor() = k_cmt0_cmcor_val;                      /* 100 Hz */
00230     *cmt0_cmcnt() = 0U;                                    /* Reset counter */
00231
00232     *icu_ir(k_vect_cmt0)  = 0U;                            /* Clear pending IRQ */
00233     *icu_ipr(k_vect_cmt0) = k_cmt0_priority;               /* Set priority */
00234     *icu_ier(k_icu_ier_cmt0) |= (uint8_t)(1U « k_icu_ier_cmt0_bit); /* Enable */
00235
00236     *cmstr0() |= k_cmstr0_ch0_bit;                         /* Start CMT0 */
00237     ___asm___ volatile("setpsw i");                        /* Global interrupt enable */
00238 }
```

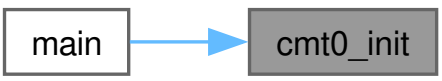
References cmstr0(), cmt0_cmcnt(), cmt0_cmcor(), cmt0_cmcr(), icu_ier(), icu_ipr(), icu_ir(), k_cmstr0_ch0_bit, k_cmt0_cmcor_val, k_cmt0_cmcr_val, k_cmt0_priority, k_icu_ier_cmt0, k_icu_ier_cmt0_bit, and k_vect_cmt0.

Referenced by main().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.4.9 cmt0_isr()

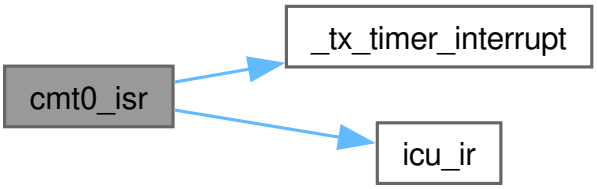
```
void cmt0_isr (
                void )
```

Definition at line 216 of file main.c.

```
00217 {
00218     *icu_ir(k_vect_cmt0) = 0U;   /* Clear interrupt request flag */
00219     _tx_timer_interrupt();       /* Drive ThreadX system clock */
00220 }
```

References _tx_timer_interrupt(), icu_ir(), and k_vect_cmt0.

Here is the call graph for this function:



4.3.4.10 flash_task_entry()

```
void flash_task_entry (
                ULONG arg)    [static]
```

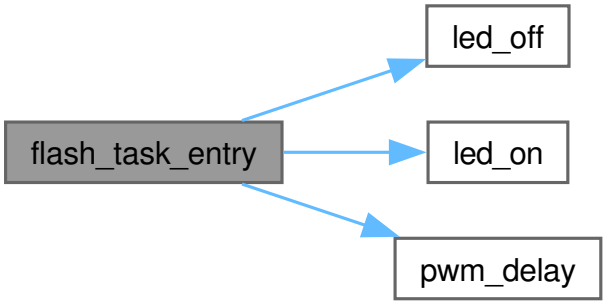
Definition at line 317 of file main.c.

```
00318 {
00319     (void)arg;
00320
00321     for (;;) {
00322         uint32_t f;
00323         uint8_t  i;
00324
00325         /* Wait for breathe_task to complete one full cycle */
00326         (void)tx_semaphore_get(&s_cycle_done_sem, TX_WAIT_FOREVER);
00327
00328         /* Wave: light each LED one at a time in sequence, k_flash_count passes.
00329          * breathe_task is blocked on s_flash_done_sem during this window,
00330          * so busy-waiting here is safe and avoids any ThreadX sleep dependency. */
00331         for (f = 0U; f < k_flash_count; f++) {
00332             for (i = 0U; i < k_num_leds; i++) {
00333                 led_on(&s_leds[i]);
00334                 pwm_delay(k_flash_on_iters);
00335                 led_off(&s_leds[i]);
00336                 pwm_delay(k_flash_off_iters);
00337             }
00338         }
00339
00340         /* Tell breathe_task it can start the next cycle */
00341         (void)tx_semaphore_put(&s_flash_done_sem);
00342     }
00343 }
```

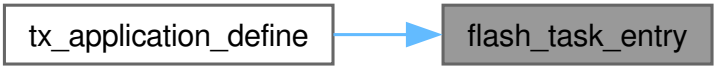
References k_flash_count, k_flash_off_iters, k_flash_on_iters, k_num_leds, led_off(), led_on(), pwm_delay(), s_cycle_done_sem, s_flash_done_sem, and s_leds.

Referenced by tx_application_define().

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.4.11 icu_ier()

volatile uint8_t * icu_ier (
uint8_t idx) [inline], [static]

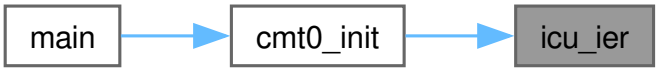
Definition at line 202 of file main.c.

```
00203 {
00204     return (volatile uint8_t *)(k_icu_ier_base + idx);
00205 }
```

References k_icu_ier_base.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.12 icu_ipr()

volatile uint8_t * icu_ipr (
uint8_t vect) [inline], [static]

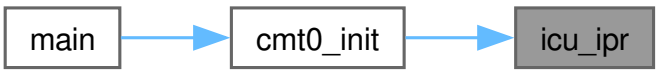
Definition at line 206 of file main.c.

```
00207 {
00208     return (volatile uint8_t *)(k_icu_ipr_base + vect);
00209 }
```

References k_icu_ipr_base.

Referenced by cmt0_init().

Here is the caller graph for this function:



4.3.4.13 icu_ir()

volatile uint8_t * icu_ir (
uint8_t vect) [inline], [static]

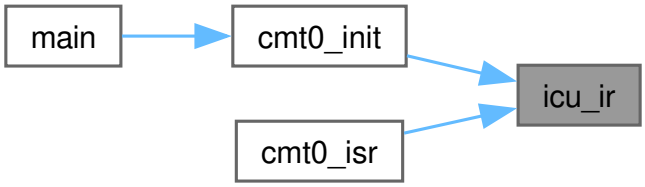
Definition at line 198 of file main.c.

```
00199 {
00200     return (volatile uint8_t *)(k_icu_ir_base + vect);
00201 }
```

References k_icu_ir_base.

Referenced by cmt0_init(), and cmt0_isr().

Here is the caller graph for this function:



4.3.4.14 led_off()

void led_off (
const led_t * led) [inline], [static]

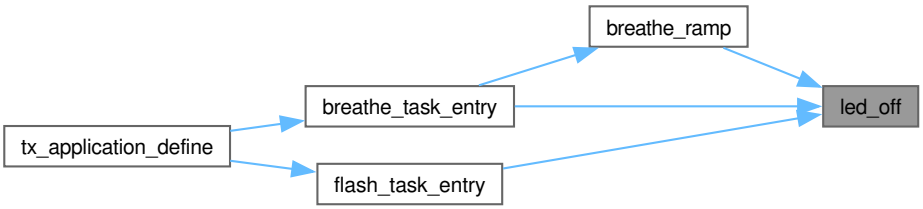
Definition at line 248 of file main.c.

```
00249 {
00250     *led->podr &= (uint8_t)~led->mask;
00251 }
```

References led_t::mask, and led_t::podr.

Referenced by breathe_ramp(), breathe_task_entry(), and flash_task_entry().

Here is the caller graph for this function:



4.3.4.15 led_on()

void led_on (
const led_t * led) [inline], [static]

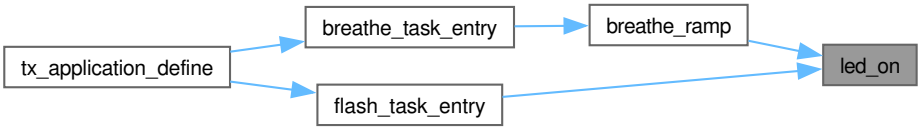
Definition at line 243 of file main.c.

```
00244 {
00245     *led->podr |= led->mask;
00246 }
```

References led_t::mask, and led_t::podr.

Referenced by breathe_ramp(), and flash_task_entry().

Here is the caller graph for this function:



4.3.4.16 main()

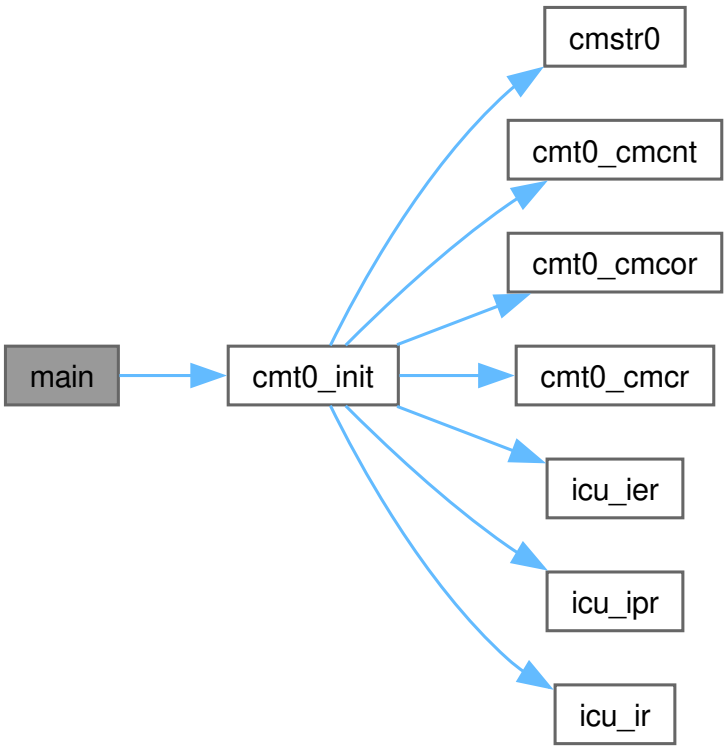
int main (
void)

Definition at line 389 of file main.c.

```
00390 {
00391     cmt0_init();
00392     tx_kernel_enter(); /* Never returns */
00393     return 0;
00394 }
```

References cmt0_init().

Here is the call graph for this function:



4.3.4.17 port7_pdr()

volatile uint8_t * port7_pdr (
void) [inline], [static]

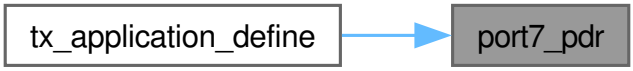
Definition at line 150 of file main.c.

```
00151 {
00152     return (volatile uint8_t *)k_port7_pdr_addr;
00153 }
```

References k_port7_pdr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.18 port7_podr()

volatile uint8_t * port7_podr (
void) [inline], [static]

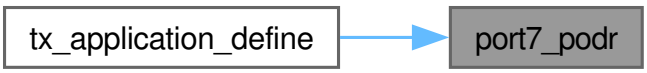
Definition at line 154 of file main.c.

```
00155 {
00156     return (volatile uint8_t *)k_port7_podr_addr;
00157 }
```

References k_port7_podr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.19 porta_pdr()

volatile uint8_t * porta_pdr (
void) [inline], [static]

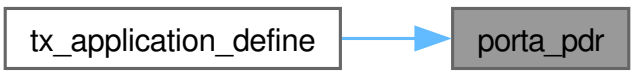
Definition at line 158 of file main.c.

```
00159 {
00160     return (volatile uint8_t *)k_porta_pdr_addr;
00161 }
```

References k_porta_pdr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.20 porta_podr()

```
volatile uint8_t * porta_podr (
                void ) [inline], [static]
```

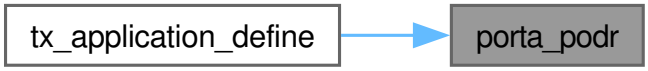
Definition at line 162 of file main.c.

```
00163 {
00164     return (volatile uint8_t *)k_porta_podr_addr;
00165 }
```

References k_porta_podr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.21 portb_pdr()

```
volatile uint8_t * portb_pdr (
                void ) [inline], [static]
```

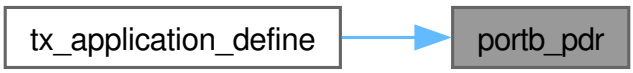
Definition at line 166 of file main.c.

```
00167 {
00168     return (volatile uint8_t *)k_portb_pdr_addr;
00169 }
```

References k_portb_pdr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.22 portb_podr()

```
volatile uint8_t * portb_podr (
                void ) [inline], [static]
```

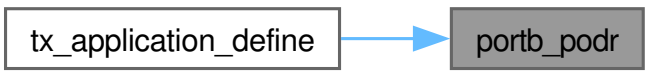
Definition at line 170 of file main.c.

```
00171 {
00172     return (volatile uint8_t *)k_portb_podr_addr;
00173 }
```

References k_portb_podr_addr.

Referenced by tx_application_define().

Here is the caller graph for this function:



4.3.4.23 pwm_delay()

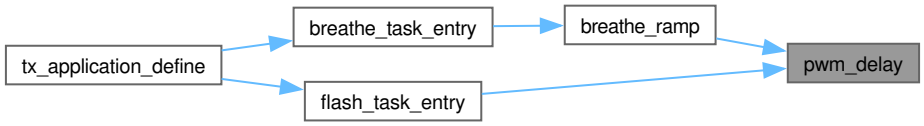
```
void pwm_delay (
                uint32_t count)   [static]
```

Definition at line 257 of file main.c.

```
00258 {
00259     volatile uint32_t i;
00260     for (i = 0U; i < count; i++) {
00261         __asm__ __volatile__("nop");
00262     }
00263 }
```

Referenced by breathe_ramp(), and flash_task_entry().

Here is the caller graph for this function:



4.3.4.24 tx_application_define()

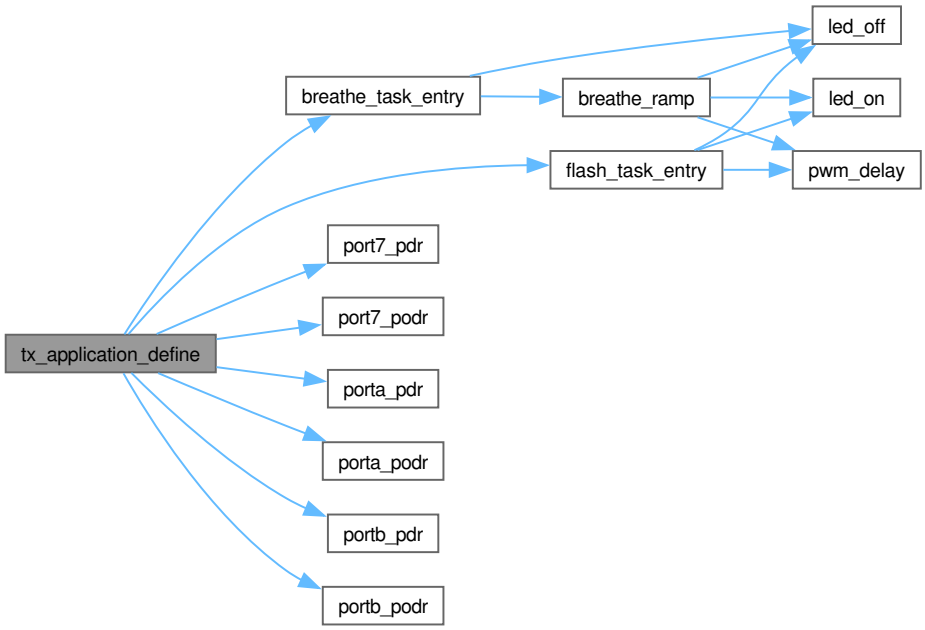
```
void tx_application_define (
                void * first_unused_memory)
```

Definition at line 349 of file main.c.

```
00350 {
00351     (void)first_unused_memory;
00352
00353     /* Semaphores: initial count 0 (both tasks wait until signalled) */
00354     (void)tx_semaphore_create(&s_cycle_done_sem, "cycle_done", 0U);
00355     (void)tx_semaphore_create(&s_flash_done_sem, "flash_done", 1U);
00356
00357     /* Configure LED pins as outputs (PDR bit = 1).
00358      * After reset PMR = 0x00 (all GPIO), no MPC unlock needed. */
00359     *porta_pdr() |= (uint8_t)k_pin_pa7;
00360     *port7_pdr() |= (uint8_t)(k_pin_p71 | k_pin_p72);
00361     *portb_pdr() |= (uint8_t)(k_pin_pb0 | k_pin_pb1 | k_pin_pb2);
00362
00363     /* Build LED table -- order defines breathing sequence */
00364     s_leds[0].podr = porta_podr(); s_leds[0].mask = (uint8_t)k_pin_pa7;
00365     s_leds[1].podr = portb_podr(); s_leds[1].mask = (uint8_t)k_pin_pb0;
00366     s_leds[2].podr = port7_podr(); s_leds[2].mask = (uint8_t)k_pin_p71;
00367     s_leds[3].podr = port7_podr(); s_leds[3].mask = (uint8_t)k_pin_p72;
00368     s_leds[4].podr = portb_podr(); s_leds[4].mask = (uint8_t)k_pin_pb1;
00369     s_leds[5].podr = portb_podr(); s_leds[5].mask = (uint8_t)k_pin_pb2;
00370
00371     /* Create breathe_task */
00372     (void)tx_thread_create(&s_breathe_tcb, "breathe",
00373                            breathe_task_entry, 0U,
00374                            s_breathe_stack, k_breathe_stack_sz,
00375                            k_breathe_priority, k_breathe_priority,
00376                            TX_NO_TIME_SLICE, TX_AUTO_START);
00377
00378     /* Create flash_task */
00379     (void)tx_thread_create(&s_flash_tcb, "flash",
00380                            flash_task_entry, 0U,
00381                            s_flash_stack, k_flash_stack_sz,
00382                            k_flash_priority, k_flash_priority,
00383                            TX_NO_TIME_SLICE, TX_AUTO_START);
00384 }
```

References breathe_task_entry(), flash_task_entry(), k_breathe_priority, k_breathe_stack_sz, k_flash_priority, k_flash_stack_sz, k_pin_p71, k_pin_p72, k_pin_pa7, k_pin_pb0, k_pin_pb1, k_pin_pb2, port7_pdr(), port7_podr(), porta_pdr(), porta_podr(), portb_pdr(), portb_podr(), s_breathe_stack, s_breathe_tcb, s_cycle_done_sem, s_flash_done_sem, s_flash_stack, s_flash_tcb, and s_leds.

Here is the call graph for this function:



4.3.5 Variable Documentation

4.3.5.1 s_breathe_stack

uint8_t s_breathe_stack[k_breathe_stack_sz] [static]

Definition at line 133 of file main.c.

Referenced by tx_application_define().

4.3.5.2 s_breathe_tcb

TX_THREAD s_breathe_tcb [static]

Definition at line 128 of file main.c.

Referenced by tx_application_define().

4.3.5.3 s_cycle_done_sem

TX_SEMAPHORE s_cycle_done_sem [static]

Posted by breathe_task, waited by flash_task

Definition at line 130 of file main.c.

Referenced by breathe_task_entry(), flash_task_entry(), and tx_application_define().

4.3.5.4 s_flash_done_sem

TX_SEMAPHORE s_flash_done_sem [static]

Posted by flash_task, waited by breathe_task

Definition at line 131 of file main.c.

Referenced by breathe_task_entry(), flash_task_entry(), and tx_application_define().

4.3.5.5 s_flash_stack

uint8_t s_flash_stack[k_flash_stack_sz] [static]

Definition at line 134 of file main.c.

Referenced by tx_application_define().

4.3.5.6 s_flash_tcb

TX_THREAD s_flash_tcb [static]

Definition at line 129 of file main.c.

Referenced by tx_application_define().

4.3.5.7 s_leds

led_t s_leds[k_num_leds] [static]

Definition at line 145 of file main.c.

Referenced by breathe_task_entry(), flash_task_entry(), and tx_application_define().

4.4 main.c

Go to the documentation of this file.

```
00001 /**
00002  * @file main.c
00003  * @brief RX72N multi-LED breathing cycle using ThreadX RTOS.
00004  *
00005  * @details
00006  * Two ThreadX threads drive the breathing animation:
00007  *
00008  *   breathe_task -- cycles through all 6 LEDs sequentially, applying a
00009  *                   software-PWM breathing effect to each one in turn.
00010  *                   After completing a full cycle it posts a semaphore.
00011  *
00012  *   flash_task   -- waits on the semaphore, runs a wave sequence: each LED
00013  *                   lights up one at a time in order (k_flash_count passes),
00014  *                   then signals breathe_task to start the next cycle.
00015  *
00016  * Threading replaces the single infinite loop of the bare-metal version.
00017  * Busy-wait delays are used for both PWM and wave timing; the 100 Hz tick is
00018  * too coarse for smooth dimming and tx_thread_sleep is not used.
00019  *
00020  * Clock: LOCO default at 240 kHz (OFS1 = 0xFFFFFFFF, factory reset state).
00021  * Tick:  CMT0 at PCLKB/8 = 30 kHz, CMCOR = 299 -> 100 Hz (10 ms/tick).
```

```
00022  *
00023  * Port register addresses: RX72N Hardware Manual r01uh0824ej0111, Ch 22.
00024  *   PDR  base = 0x0008C000 + port_index
00025  *   PODR base = 0x0008C020 + port_index
00026  *   Port indices: 7=0x07, A=0x0A, B=0x0B
00027  *
00028  * ICU and CMT0 addresses: RX72N Hardware Manual, Ch 13/15.
00029  *
00030  * SPDX-License-Identifier: MIT
00031  * @copyright Copyright (c) 2026 Locked Inc.
00032  */
00033
00034 #include <stdint.h>
00035 #include "tx_api.h"
00036
00037 /*
       ============================================================================
00038  * Hardware register addresses
00039  * uintptr_t is mandatory: on RX72N target = uint32_t, on x86-64 host =
00040  * uint64_t.  Using uint32_t directly silently truncates on the host.
00041  *
       ============================================================================
       */
00042 typedef enum : uintptr_t {
00043     /* GPIO */
00044     k_port7_pdr_addr  = 0x0008C007U, /**< PORT7 direction register */
00045     k_port7_podr_addr = 0x0008C027U, /**< PORT7 output data register */
00046     k_porta_pdr_addr  = 0x0008C00AU, /**< PORTA direction register */
00047     k_porta_podr_addr = 0x0008C02AU, /**< PORTA output data register */
00048     k_portb_pdr_addr  = 0x0008C00BU, /**< PORTB direction register */
00049     k_portb_podr_addr = 0x0008C02BU, /**< PORTB output data register */
00050
00051     /* CMT0 (compare match timer 0) */
00052     k_cmstr0_addr = 0x00088000U, /**< CMT start register (shared CH0/CH1) */
00053     k_cmt0_cmcr   = 0x00088002U, /**< CMT0 control register */
00054     k_cmt0_cmcnt  = 0x00088004U, /**< CMT0 counter */
00055     k_cmt0_cmcor  = 0x00088006U, /**< CMT0 compare match register */
00056
00057     /* ICU (interrupt controller) */
00058     k_icu_ir_base  = 0x00087000U, /**< IR[] base (IR[n] = base + n) */
00059     k_icu_ier_base = 0x00087200U, /**< IER[] base (IER[n/8] bit n%8) */
00060     k_icu_ipr_base = 0x00087300U, /**< IPR[] base (IPR[n] = base + n) */
00061 } hw_addr_t;
00062
00063 /*
       ============================================================================
00064  * Pin bit masks (one enum per port to avoid duplicate-value collisions)
00065  *
       ============================================================================
       */
00066 typedef enum : uint8_t {
00067     k_pin_pa7 = (uint8_t)(1U << 7U), /**< PORTA pin 7 */
00068 } porta_pins_t;
00069
00070 typedef enum : uint8_t {
00071     k_pin_p71 = (uint8_t)(1U << 1U), /**< PORT7 pin 1 */
00072     k_pin_p72 = (uint8_t)(1U << 2U), /**< PORT7 pin 2 */
00073 } port7_pins_t;
00074
00075 typedef enum : uint8_t {
00076     k_pin_pb0 = (uint8_t)(1U << 0U), /**< PORTB pin 0 */
00077     k_pin_pb1 = (uint8_t)(1U << 1U), /**< PORTB pin 1 */
00078     k_pin_pb2 = (uint8_t)(1U << 2U), /**< PORTB pin 2 */
00079 } portb_pins_t;
00080
00081 /*
       ============================================================================
00082  * CMT0 configuration constants (LOCO = 240 kHz)
00083  *   PCLKB/8 = 30 000 Hz timer clock.
00084  *   CMCOR = 30000/100 - 1 = 299  ->  100 Hz tick (10 ms/tick).
00085  *   CMCR  = 0x40  ->  CMIE=1 (interrupt enable), CKS=00 (PCLKB/8).
00086  *
       ============================================================================
       */
00087 typedef enum : uint16_t {
00088     k_cmt0_cmcor_val  = 299U,  /**< Compare match for 100 Hz at LOCO/8 */
00089     k_cmt0_cmcr_val   = 0x40U, /**< CMIE=1, CKS=00 (PCLKB/8) */
00090     k_cmstr0_ch0_bit  = 0x01U, /**< CMSTR0 bit 0 = CMT0 start/stop */
00091 } cmt0_cfg_t;
00092
00093 typedef enum : uint8_t {
00094     k_vect_cmt0       = 28U,  /**< CMT0 CMI0 interrupt vector number */
00095     k_cmt0_priority   = 3U,   /**< ICU interrupt priority for CMT0 */
00096     k_icu_ier_cmt0    = 3U,   /**< IER register index for vector 28 (28/8) */
00097     k_icu_ier_cmt0_bit = 4U,  /**< Bit position within IER[3] (28%8) */
00098 } icu_cmt0_cfg_t;
00099
```

```
00100 /*
      ============================================================================
00101  * Breathing / flash timing
00102  *   Each delay() iteration ~= 50 us at LOCO 240 kHz (same as bare-metal).
00103  *   PWM period = k_max_bright iters = 50 * 50 us = 2.5 ms -> 400 Hz PWM.
00104  *   tx_thread_sleep() granularity = 10 ms/tick.
00105  *
      ============================================================================
      */
00106 typedef enum : uint32_t {
00107     k_max_bright       = 50U,    /**< PWM duty-cycle upper bound */
00108     k_reps_per_step    = 5U,     /**< PWM cycles per brightness step */
00109     k_flash_count      = 3U,     /**< Wave passes at cycle end */
00110     k_flash_on_iters   = 4000U,  /**< Wave LED ON busy-wait iters (~200 ms at LOCO 240 kHz) */
00111     k_flash_off_iters  = 2000U,  /**< Wave LED OFF gap busy-wait iters (~100 ms at LOCO 240 kHz) */
00112 } breathe_params_t;
00113
00114 /*
      ============================================================================
00115  * ThreadX objects
00116  *
      ============================================================================
      */
00117 typedef enum : uint8_t {
00118     k_num_leds         = 6U,  /**< LEDs in the breathing sequence */
00119     k_breathe_priority = 10U, /**< breathe_task ThreadX priority */
00120     k_flash_priority   = 9U,  /**< flash_task ThreadX priority (runs first) */
00121 } rtos_prio_t;
00122
00123 typedef enum : uint16_t {
00124     k_breathe_stack_sz = 512U, /**< breathe_task stack in bytes */
00125     k_flash_stack_sz   = 256U, /**< flash_task stack in bytes */
00126 } rtos_stack_t;
00127
00128 static TX_THREAD     s_breathe_tcb;
00129 static TX_THREAD     s_flash_tcb;
00130 static TX_SEMAPHORE s_cycle_done_sem;  /**< Posted by breathe_task, waited by flash_task */
00131 static TX_SEMAPHORE s_flash_done_sem;  /**< Posted by flash_task, waited by breathe_task */
00132
00133 static uint8_t s_breathe_stack[k_breathe_stack_sz];
00134 static uint8_t s_flash_stack[k_flash_stack_sz];
00135
00136 /*
      ============================================================================
00137  * LED descriptor
00138  *
      ============================================================================
      */
00139 typedef struct {
00140     volatile uint8_t *podr; /**< Port output data register */
00141     uint8_t           mask; /**< Bit mask for this LED's pin */
00142 } led_t;
00143
00144 /* Shared LED table (read-only after init in tx_application_define) */
00145 static led_t s_leds[k_num_leds];
00146
00147 /*
      ============================================================================
00148  * Inline port register accessors
00149  *
      ============================================================================
      */
00150 static inline volatile uint8_t *port7_pdr(void)
00151 {
00152     return (volatile uint8_t *)k_port7_pdr_addr;
00153 }
00154 static inline volatile uint8_t *port7_podr(void)
00155 {
00156     return (volatile uint8_t *)k_port7_podr_addr;
00157 }
00158 static inline volatile uint8_t *porta_pdr(void)
00159 {
00160     return (volatile uint8_t *)k_porta_pdr_addr;
00161 }
00162 static inline volatile uint8_t *porta_podr(void)
00163 {
00164     return (volatile uint8_t *)k_porta_podr_addr;
00165 }
00166 static inline volatile uint8_t *portb_pdr(void)
00167 {
00168     return (volatile uint8_t *)k_portb_pdr_addr;
00169 }
00170 static inline volatile uint8_t *portb_podr(void)
00171 {
00172     return (volatile uint8_t *)k_portb_podr_addr;
00173 }
00174
```

```
00175 /*
      ============================================================================
00176  * CMT0 register accessors
00177  *
      ============================================================================
      */
00178 static inline volatile uint16_t *cmstr0(void)
00179 {
00180     return (volatile uint16_t *)k_cmstr0_addr;
00181 }
00182 static inline volatile uint16_t *cmt0_cmcr(void)
00183 {
00184     return (volatile uint16_t *)k_cmt0_cmcr;
00185 }
00186 static inline volatile uint16_t *cmt0_cmcnt(void)
00187 {
00188     return (volatile uint16_t *)k_cmt0_cmcnt;
00189 }
00190 static inline volatile uint16_t *cmt0_cmcor(void)
00191 {
00192     return (volatile uint16_t *)k_cmt0_cmcor;
00193 }
00194
00195 /*
      ============================================================================
00196  * ICU register accessors
00197  *
      ============================================================================
      */
00198 static inline volatile uint8_t *icu_ir(uint8_t vect)
00199 {
00200     return (volatile uint8_t *)(k_icu_ir_base + vect);
00201 }
00202 static inline volatile uint8_t *icu_ier(uint8_t idx)
00203 {
00204     return (volatile uint8_t *)(k_icu_ier_base + idx);
00205 }
00206 static inline volatile uint8_t *icu_ipr(uint8_t vect)
00207 {
00208     return (volatile uint8_t *)(k_icu_ipr_base + vect);
00209 }
00210
00211 /*
      ============================================================================
00212  * ThreadX tick source -- CMT0 ISR
00213  *
      ============================================================================
      */
00214 extern void _tx_timer_interrupt(void);
00215
00216 void __attribute__((interrupt)) cmt0_isr(void)
00217 {
00218     *icu_ir(k_vect_cmt0) = 0U;   /* Clear interrupt request flag */
00219     _tx_timer_interrupt();       /* Drive ThreadX system clock */
00220 }
00221
00222 /*
      ============================================================================
00223  * CMT0 initialisation (called before tx_kernel_enter)
00224  *
      ============================================================================
      */
00225 static void cmt0_init(void)
00226 {
00227     *cmstr0()    &= (uint16_t)~(uint16_t)k_cmstr0_ch0_bit; /* Stop CMT0 */
00228     *cmt0_cmcr() = k_cmt0_cmcr_val;                        /* PCLKB/8, IRQ enabled */
00229     *cmt0_cmcor() = k_cmt0_cmcor_val;                      /* 100 Hz */
00230     *cmt0_cmcnt() = 0U;                                    /* Reset counter */
00231
00232     *icu_ir(k_vect_cmt0)  = 0U;                            /* Clear pending IRQ */
00233     *icu_ipr(k_vect_cmt0) = k_cmt0_priority;               /* Set priority */
00234     *icu_ier(k_icu_ier_cmt0) |= (uint8_t)(1U << k_icu_ier_cmt0_bit); /* Enable */
00235
00236     *cmstr0() |= k_cmstr0_ch0_bit;                         /* Start CMT0 */
00237     __asm__ volatile("setpsw i");                          /* Global interrupt enable */
00238 }
00239
00240 /*
      ============================================================================
00241  * LED helpers
00242  *
      ============================================================================
      */
00243 static inline void led_on(const led_t *led)
00244 {
00245     *led->podr |= led->mask;
00246 }
```

```
00247
00248 static inline void led_off(const led_t *led)
00249 {
00250     *led->podr &= (uint8_t)~led->mask;
00251 }
00252
00253 /*
      ============================================================================
00254  * Busy-wait delay (used for sub-tick software PWM)
00255  * One iteration ~= 50 us at LOCO 240 kHz.
00256  *
      ============================================================================
      */
00257 static void pwm_delay(uint32_t count)
00258 {
00259     volatile uint32_t i;
00260     for (i = 0U; i < count; i++) {
00261         __asm__ __volatile__("nop");
00262     }
00263 }
00264
00265 /*
      ============================================================================
00266  * Software PWM ramp -- one direction of the breathe animation.
00267  *
      ============================================================================
      */
00268 static void breathe_ramp(const led_t *led, uint32_t invert)
00269 {
00270     uint32_t b;
00271     uint32_t rep;
00272     uint32_t on_time;
00273     uint32_t off_time;
00274
00275     for (b = 0U; b <= k_max_bright; b++) {
00276         on_time  = (invert == 0U) ? b : (k_max_bright - b);
00277         off_time = k_max_bright - on_time;
00278         for (rep = 0U; rep < k_reps_per_step; rep++) {
00279             if (on_time > 0U) {
00280                 led_on(led);
00281                 pwm_delay(on_time);
00282             }
00283             led_off(led);
00284             if (off_time > 0U) {
00285                 pwm_delay(off_time);
00286             }
00287         }
00288     }
00289 }
00290
00291 /*
      ============================================================================
00292  * breathe_task -- cycles through all LEDs, posts semaphore when done.
00293  *
      ============================================================================
      */
00294 static void breathe_task_entry(ULONG arg)
00295 {
00296     (void)arg;
00297
00298     for (;;) {
00299         uint8_t i;
00300         for (i = 0U; i < k_num_leds; i++) {
00301             breathe_ramp(&s_leds[i], 0U); /* fade in  */
00302             breathe_ramp(&s_leds[i], 1U); /* fade out */
00303             led_off(&s_leds[i]);
00304         }
00305
00306         /* Signal flash_task that one full cycle is complete */
00307         (void)tx_semaphore_put(&s_cycle_done_sem);
00308
00309         /* Wait for flash_task to finish before starting the next cycle */
00310         (void)tx_semaphore_get(&s_flash_done_sem, TX_WAIT_FOREVER);
00311     }
00312 }
00313
00314 /*
      ============================================================================
00315  * flash_task -- waits for cycle complete, flashes all LEDs, signals back.
00316  *
      ============================================================================
      */
00317 static void flash_task_entry(ULONG arg)
00318 {
00319     (void)arg;
00320
00321     for (;;) {
```

```
00322         uint32_t f;
00323         uint8_t  i;
00324
00325         /* Wait for breathe_task to complete one full cycle */
00326         (void)tx_semaphore_get(&s_cycle_done_sem, TX_WAIT_FOREVER);
00327
00328         /* Wave: light each LED one at a time in sequence, k_flash_count passes.
00329          * breathe_task is blocked on s_flash_done_sem during this window,
00330          * so busy-waiting here is safe and avoids any ThreadX sleep dependency. */
00331         for (f = 0U; f < k_flash_count; f++) {
00332             for (i = 0U; i < k_num_leds; i++) {
00333                 led_on(&s_leds[i]);
00334                 pwm_delay(k_flash_on_iters);
00335                 led_off(&s_leds[i]);
00336                 pwm_delay(k_flash_off_iters);
00337             }
00338         }
00339
00340         /* Tell breathe_task it can start the next cycle */
00341         (void)tx_semaphore_put(&s_flash_done_sem);
00342     }
00343 }
00344
00345 /*
     ==========================================================================
00346  * tx_application_define -- called by ThreadX after tx_kernel_enter().
00347  * Creates semaphores, configures GPIO, and starts both threads.
00348  *
     ==========================================================================
     */
00349 void tx_application_define(void *first_unused_memory)
00350 {
00351     (void)first_unused_memory;
00352
00353     /* Semaphores: initial count 0 (both tasks wait until signalled) */
00354     (void)tx_semaphore_create(&s_cycle_done_sem, "cycle_done", 0U);
00355     (void)tx_semaphore_create(&s_flash_done_sem, "flash_done", 1U);
00356
00357     /* Configure LED pins as outputs (PDR bit = 1).
00358      * After reset PMR = 0x00 (all GPIO), no MPC unlock needed. */
00359     *porta_pdr() |= (uint8_t)k_pin_pa7;
00360     *port7_pdr() |= (uint8_t)(k_pin_p71 | k_pin_p72);
00361     *portb_pdr() |= (uint8_t)(k_pin_pb0 | k_pin_pb1 | k_pin_pb2);
00362
00363     /* Build LED table -- order defines breathing sequence */
00364     s_leds[0].podr = porta_podr(); s_leds[0].mask = (uint8_t)k_pin_pa7;
00365     s_leds[1].podr = portb_podr(); s_leds[1].mask = (uint8_t)k_pin_pb0;
00366     s_leds[2].podr = port7_podr(); s_leds[2].mask = (uint8_t)k_pin_p71;
00367     s_leds[3].podr = port7_podr(); s_leds[3].mask = (uint8_t)k_pin_p72;
00368     s_leds[4].podr = portb_podr(); s_leds[4].mask = (uint8_t)k_pin_pb1;
00369     s_leds[5].podr = portb_podr(); s_leds[5].mask = (uint8_t)k_pin_pb2;
00370
00371     /* Create breathe_task */
00372     (void)tx_thread_create(&s_breathe_tcb, "breathe",
00373                            breathe_task_entry, 0U,
00374                            s_breathe_stack, k_breathe_stack_sz,
00375                            k_breathe_priority, k_breathe_priority,
00376                            TX_NO_TIME_SLICE, TX_AUTO_START);
00377
00378     /* Create flash_task */
00379     (void)tx_thread_create(&s_flash_tcb, "flash",
00380                            flash_task_entry, 0U,
00381                            s_flash_stack, k_flash_stack_sz,
00382                            k_flash_priority, k_flash_priority,
00383                            TX_NO_TIME_SLICE, TX_AUTO_START);
00384 }
00385
00386 /*
     ==========================================================================
00387  * main -- initialise CMT0 tick source, then hand off to ThreadX forever.
00388  *
     ==========================================================================
     */
00389 int main(void)
00390 {
00391     cmt0_init();
00392     tx_kernel_enter(); /* Never returns */
00393     return 0;
00394 }
```